

INCM Assignment 2

Hodgkin–Huxley Dynamics and a Cav3.1 Calcium Channel

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This report summarizes the simulations in the accompanying Python notebook. The models use the classical Hodgkin–Huxley voltage convention, in which the resting potential is approximately 0 mV, and the Cav3.1 equations are converted from Channelpedia voltage coordinates as described below.

Overview

The assignment examines action-potential generation in the classical Hodgkin–Huxley (HH) model, measures how sodium conductance changes repetitive firing, and extends the model with a low-threshold Cav3.1 calcium channel to test rebound firing after hyperpolarization. Simulations use Brian2 and the `exponential_euler` method.

Part A: Classical Hodgkin–Huxley Model

Q1: Dynamics of an action potential

The HH membrane equation implemented by the model is

$$C_m \frac{dV}{dt} = I_{\text{ext}} + \bar{g}_{\text{Na}} m^3 h (E_{\text{Na}} - V) + \bar{g}_{\text{K}} n^4 (E_{\text{K}} - V) + g_{\text{L}} (E_{\text{L}} - V). \quad (1)$$

Here m is sodium activation, h is sodium inactivation, and n is potassium activation. The notebook applies a $7.2 \mu\text{A}$ step from 10 to 12 ms, simulates for 50 ms, and uses $\Delta t = 0.01$ ms.

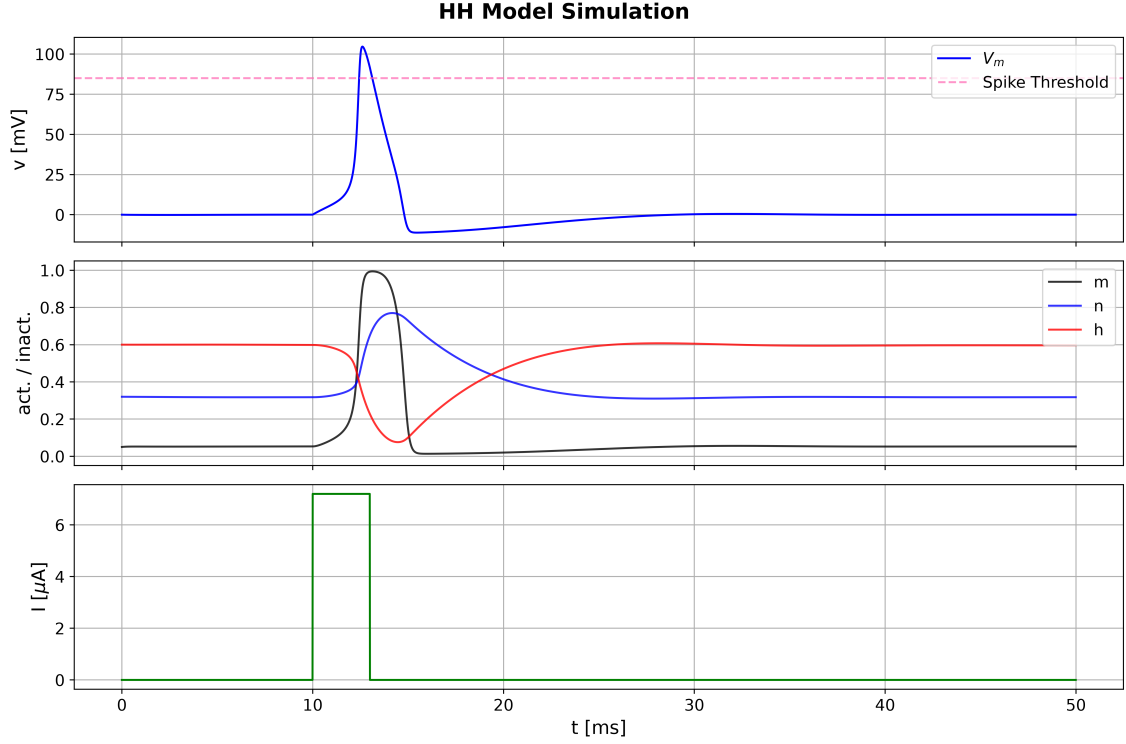


Figure 1: Classical HH response to a brief depolarizing current step.

The neuron begins near the classical HH resting state, $V \approx 0$ mV, with $m \approx 0.05$, $h \approx 0.60$, and $n \approx 0.32$. The current step rapidly increases sodium activation. The resulting inward sodium current reinforces depolarization and produces the upstroke. Sodium inactivation and delayed potassium activation then terminate the spike and produce the afterhyperpolarization. Recovery of the gates returns the membrane toward rest.

Q2: Effect of sodium conductance on excitability

The standard model uses $\bar{g}_{\text{Na}} = 120$ mS; the modified model uses $1.4 \times 120 = 168$ mS. A current step is applied from 50 to 350 ms during a 400 ms simulation. Currents are swept from 0 to $15 \mu\text{A}$ in increments of $0.1 \mu\text{A}$. Repetitive firing means at least three upward crossings of 20 mV during the injection interval. The threshold search uses $\Delta t = 0.05$ ms. The near-threshold traces are then generated at both $\Delta t = 0.05$ ms and the finer $\Delta t = 0.01$ ms to check that the qualitative comparison is not an integration-step artifact.

Table 1: Repetitive-firing thresholds obtained by the notebook search.

Model	$\bar{g}_{\text{Na}}$	I_{th}	Spikes at threshold
Standard HH	120 mS	$6.2 \mu\text{A}$	3
Modified HH	168 mS	$1.2 \mu\text{A}$	13

The measured threshold reduction is

$$\frac{6.2 - 1.2}{6.2} \times 100\% \approx 80.6\%. \quad (2)$$

Increasing $\bar{g}_{Na}$ strengthens the regenerative sodium feedback, so less external current is needed to overcome leak and potassium currents. The modified model also produces more spikes during the fixed 300 ms step, although this count is a finite-window measure rather than an asymptotic firing rate.

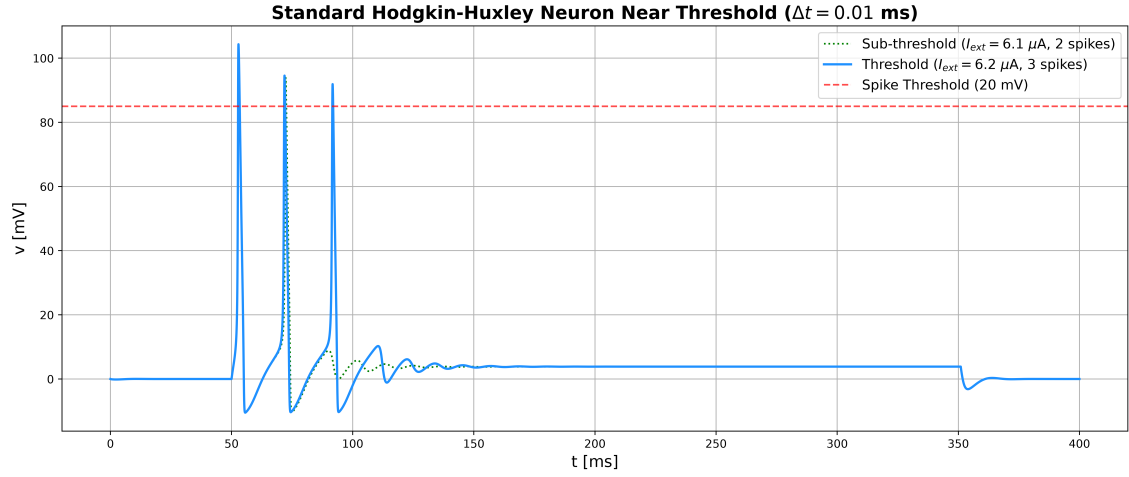


Figure 2: Standard HH: a current just below the measured repetitive-firing threshold ($6.1 \mu A$, two spikes) compared with the threshold current ($6.2 \mu A$, three spikes), using $\Delta t = 0.01$ ms.

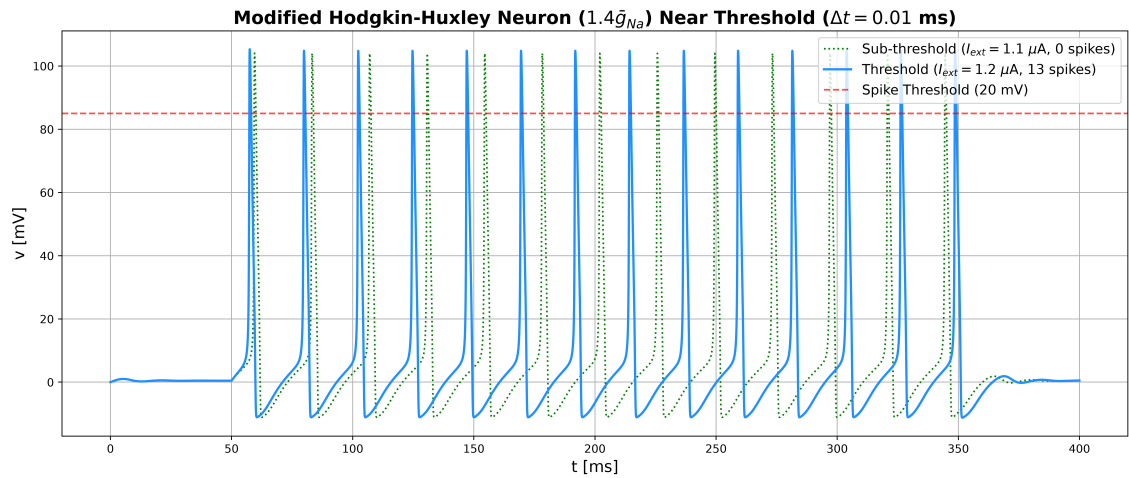


Figure 3: Modified HH: a current below the measured repetitive-firing threshold ($1.1 \mu A$, no spikes) compared with the threshold current ($1.2 \mu A$, thirteen spikes), using $\Delta t = 0.01$ ms.

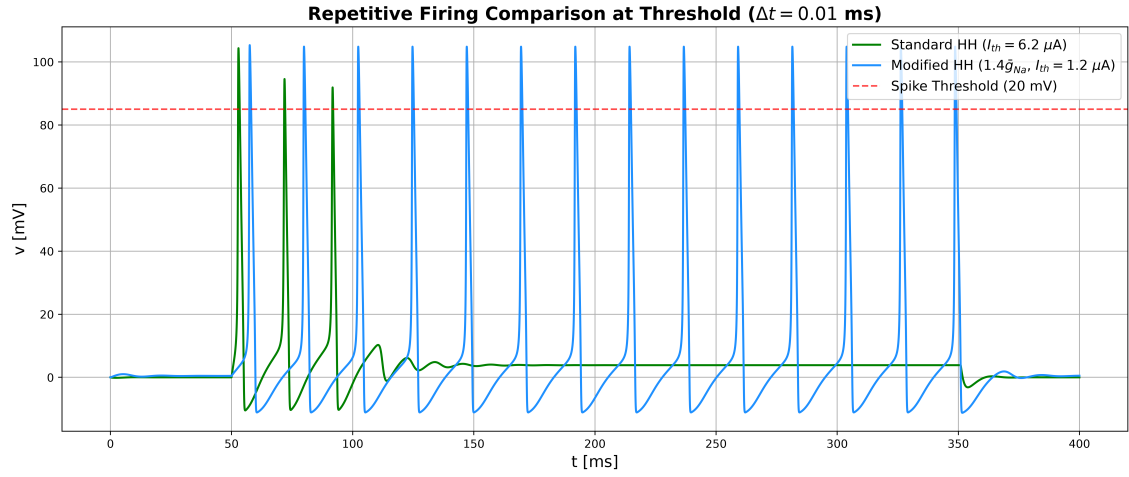


Figure 4: Direct comparison of the two models at their respective measured repetitive-firing thresholds, using $\Delta t = 0.01$ ms.

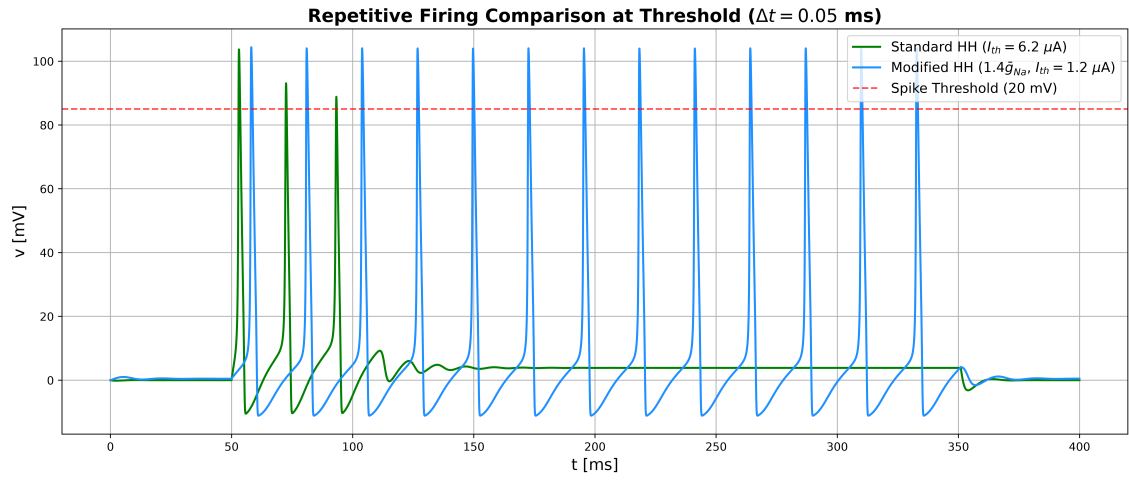


Figure 5: The same threshold comparison at the coarser search timestep, $\Delta t = 0.05$ ms.

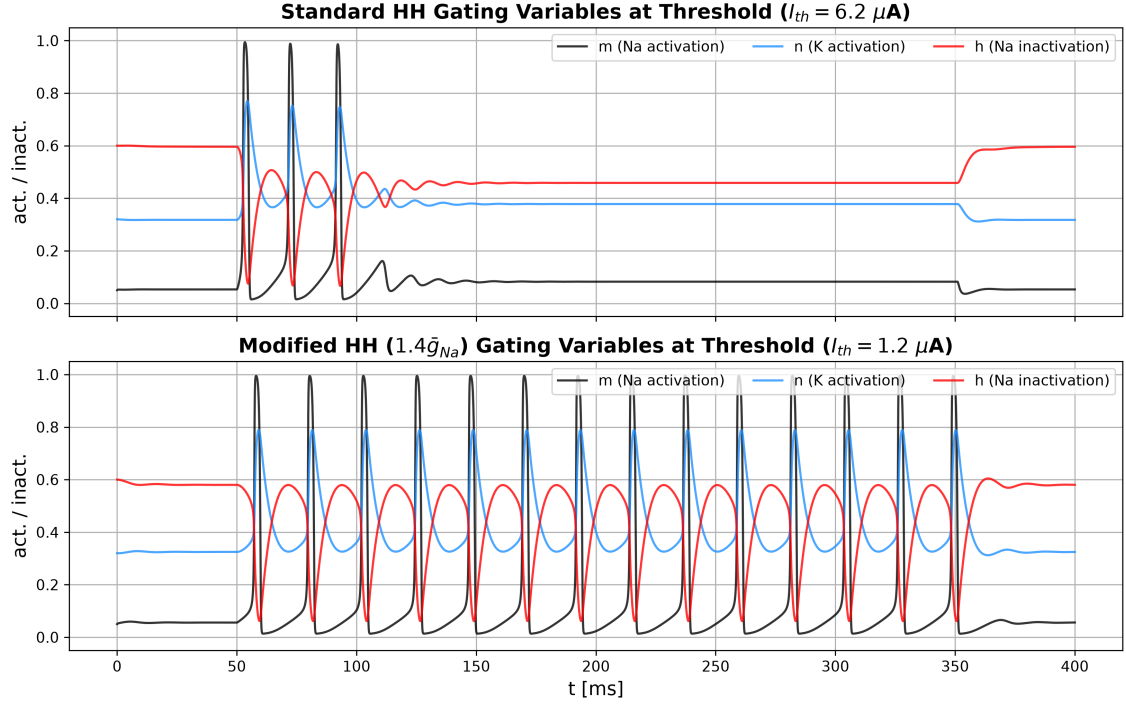


Figure 6: Standard and modified HH gating variables at their respective measured repetitive-firing thresholds.

Figures 4 and 5 show the same model comparison at the two integration steps. The 0.01 ms plots are the final high-resolution traces; the 0.05 ms plot documents the resolution used during the threshold search. The table values are defined by the search criterion, not by the visual height of the voltage traces.

Part B: Adding the Cav3.1 Calcium Channel

Q3: Channel characterization and kinetics

Cav3.1 is a low-threshold calcium channel with one activation gate m_{Ca} and one inactivation gate h_{Ca} , each with power one. The Channelpedia reversal potential is $E_{Ca,CP} = 30$ mV. The HH implementation uses $E_{Ca,HH} = 95$ mV and $V_{CP} = V_{HH} - 65$ mV.

The implemented current is

$$I_{Ca} = \bar{g}_{Ca} m_{Ca} h_{Ca} (E_{Ca} - V), \quad (3)$$

which is inward and depolarizing for $V < E_{Ca}$. The gate dynamics are

$$\frac{dm_{Ca}}{dt} = \frac{m_{\infty}(V) - m_{Ca}}{\tau_m(V)}, \quad (4)$$

$$\frac{dh_{Ca}}{dt} = \frac{h_{\infty}(V) - h_{Ca}}{\tau_h(V)}, \quad (5)$$

$$m_{\infty}(V) = \frac{1}{1 + \exp((V + 42.921064)/(-5.163208))}, \quad (6)$$

$$h_{\infty}(V) = \frac{1}{1 + \exp((V + 72.907420)/4.575763)}, \quad (7)$$

$$\tau_m(V) = \begin{cases} -0.855809 + 1.493527 \exp(-V/27.414182) \text{ ms}, & V < -10 \text{ mV}, \\ 1.0 \text{ ms}, & V \geq -10 \text{ mV}, \end{cases} \quad (8)$$

$$\tau_h(V) = 9.987873 + 0.002883 \exp(-V/5.598574) \text{ ms}. \quad (9)$$

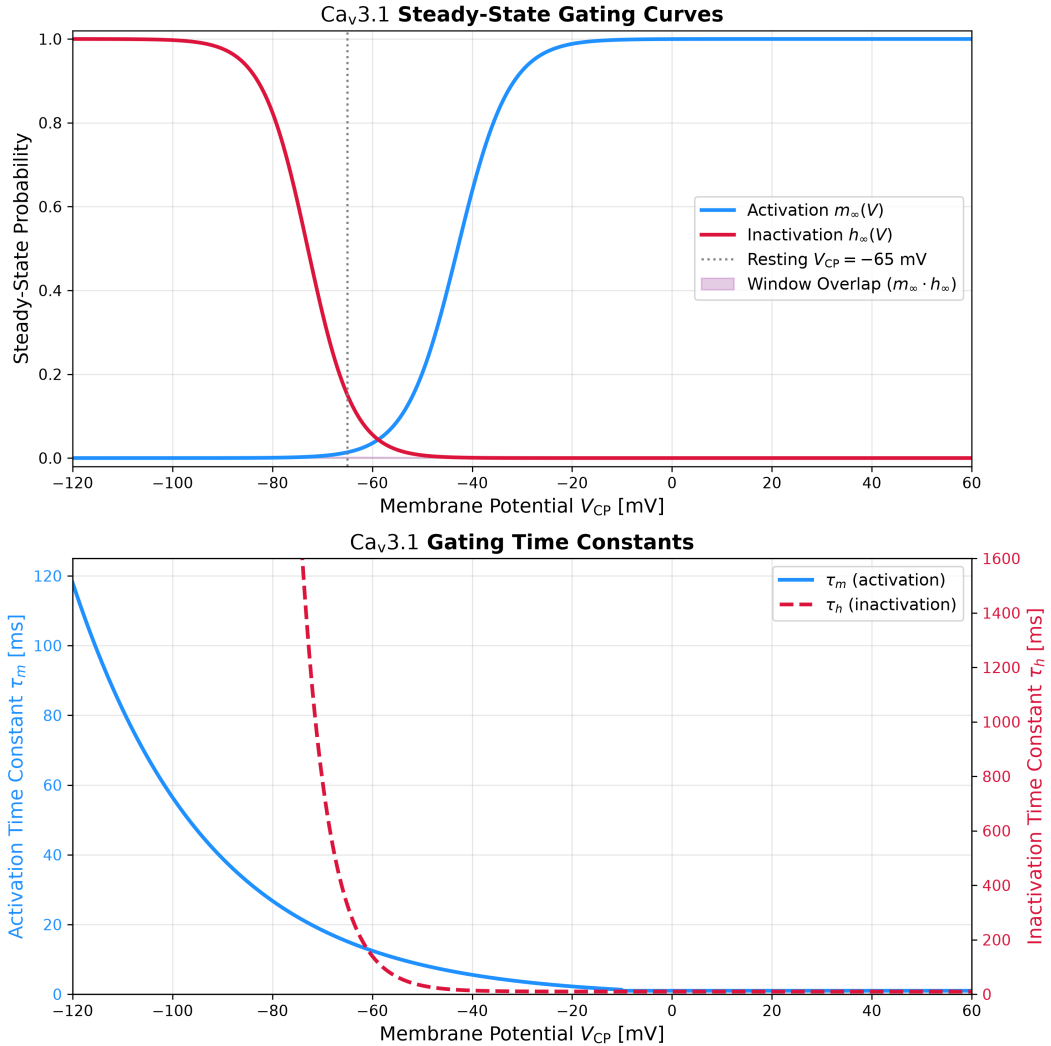


Figure 7: Cav3.1 steady-state gating variables and voltage-dependent time constants.

At hyperpolarized voltages, h_{∞} is large but m_{∞} is small: the channel is available but mostly closed. During depolarization, m_{∞} rises while h_{∞} falls. Thus, hyperpolarization

primes a transient current by removing inactivation, and release allows activation before inactivation has fully recovered. The channel is therefore most relevant during a transient excursion through approximately -55 to -35 mV in Channelpedia coordinates, where activation has begun but inactivation has not yet eliminated availability. The activation gate is faster than the inactivation gate, so a brief overlap of m_{Ca} and h_{Ca} is possible after hyperpolarization.

Q4: Rebound response after hyperpolarization

The extended model uses $\bar{g}_{Ca} = 20$ mS and applies a -2.0 μ A step from 50 to 250 ms. Both models are simulated for 350 ms with $\Delta t = 0.01$ ms.

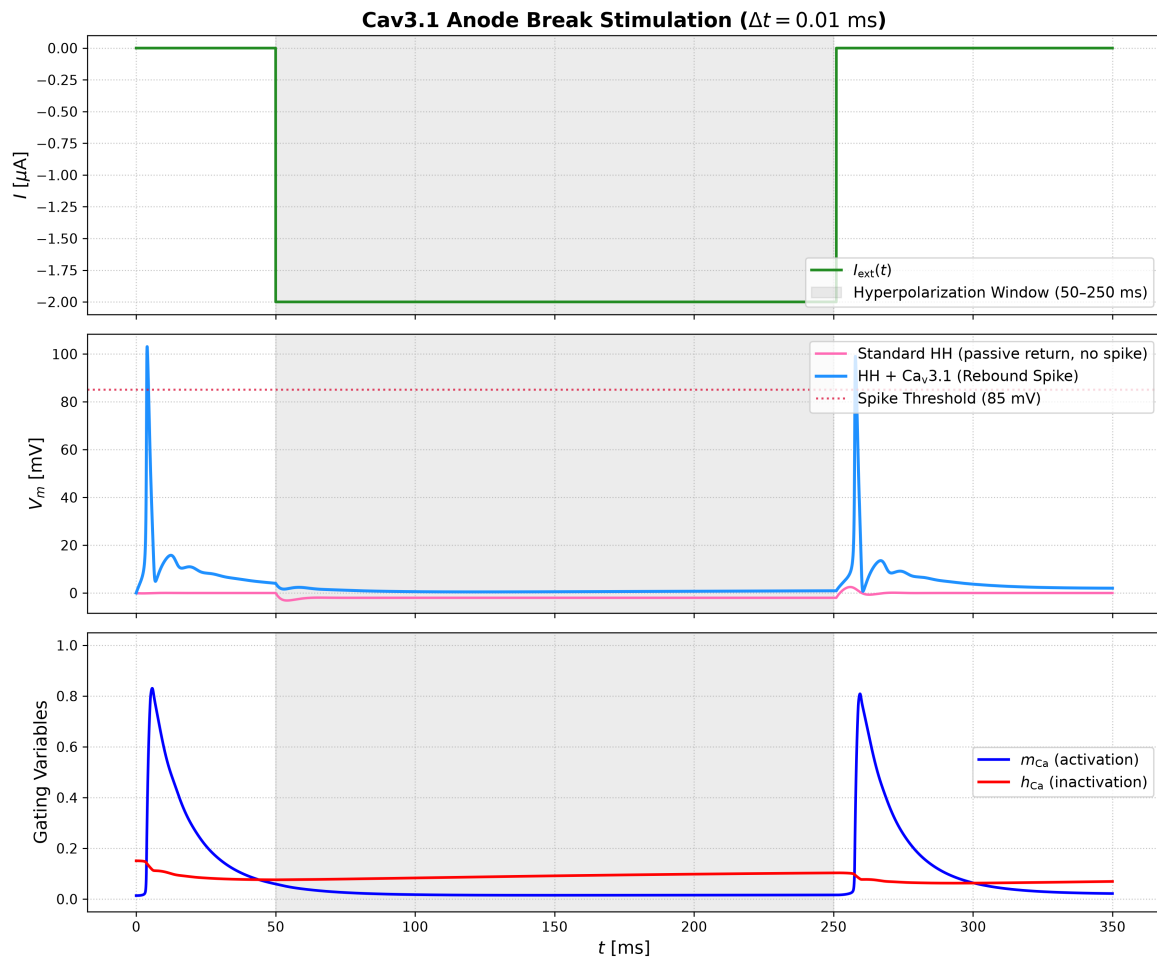


Figure 8: Anode-break stimulation: hyperpolarization primes Cav3.1, followed by a rebound spike after current removal.

During the negative-current interval, h_{Ca} increases while m_{Ca} remains small. When the current is removed at 250 ms, the membrane returns toward rest. Activation of m_{Ca} overlaps with the still-large h_{Ca} , producing an inward calcium current that depolarizes the membrane enough to recruit the standard sodium current. The standard HH comparison returns toward rest without the corresponding rebound event.

For the saved notebook run, the rebound peak is 99.02 mV at 257.84 ms, giving a peak latency of 7.84 ms. The first crossing of the analysis threshold 20 mV occurs at 257.68 ms, giving a threshold-crossing latency of 7.68 ms.

Conclusion

The simulations demonstrate two mechanisms of excitability. In the classical HH model, sodium conductance controls regenerative positive feedback and strongly changes the current required for repetitive firing. In the extended model, Cav3.1 provides a memory of prior hyperpolarization: its inactivation gate is relieved during the negative-current pulse, and release produces a transient inward calcium current. The rebound spike is therefore a direct consequence of voltage-dependent activation and inactivation kinetics.